EUNOIA JUNIOR COLLEGE
JC1 Promotional Examination 2020
General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME	

CIVICS GROUP	2	0	–			REGISTRATION NUMBER		

CHEMISTRY

9729/02

Paper 2 Structured Questions and Free Response

01 October 2020
2 hours

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Do not use paper clips, highlighters, glue or correction fluid.
Write your name, civics group, registration number on all the work you hand in.

Answer all questions in the spaces provided on this Question Paper.

Section A
You are advised to spend 1 hour 30 minutes on this section.
There are five questions in this section.
Answer **all** questions in this section.

Section B
You are advised to spend 30 minutes on this section.
There are two questions in this section. **Choose 1 out of the 2 questions**.

Write all answers in **dark blue or black pen**.
You may use an HB pencil for any diagrams or graphs.

The number of marks is given in brackets [ ] at the end of each question or part question.

A Data Booklet is provided.
The use of an approved scientific calculator is expected, where appropriate.

For Examiner's Use	
Paper 2	
Section A	
1	/10
2	/17
3	/ 8
4	/12
5	/13
Section B	
6 or 7	/20
P2 Total	/80
Paper 1	/30
Paper 4	/33

Section A

Answer **all** the questions in this section.

1 *Use of the Data Booklet is relevant to this question*.

Chromium has two naturally occurring isotopes, ^{52}Cr and ^{54}Cr. Some common oxidation states of chromium are +2 and +3.

(a) Write the full electronic configuration of Cr.

Cr: **$1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$** .... [1]

(b) Explain why the 2nd ionisation energy of Cr is lower than its 3rd ionisation energy.

Cr^+ and Cr^{2+} ions have **the same nuclear charge**, but **the number of electrons decrease such that the inter-electronic repulsion decrease**. There will be a **stronger electrostatic forces of attraction** between the nucleus and the remaining electrons, and **more energy** is needed for subsequent removal of electrons. Hence, the 2nd I.E is lower that that of the 3rd I.E. .... [2]

(c) When passed through an electric field of a certain charge, $^{52}Cr^{2+}$ ion deflects by an angle of 6º.

(i) Calculate the angle of deflection of $^{54}Cr^{3+}$ ion when passed through the same electric field.

$$\theta \propto \frac{q}{m}$$

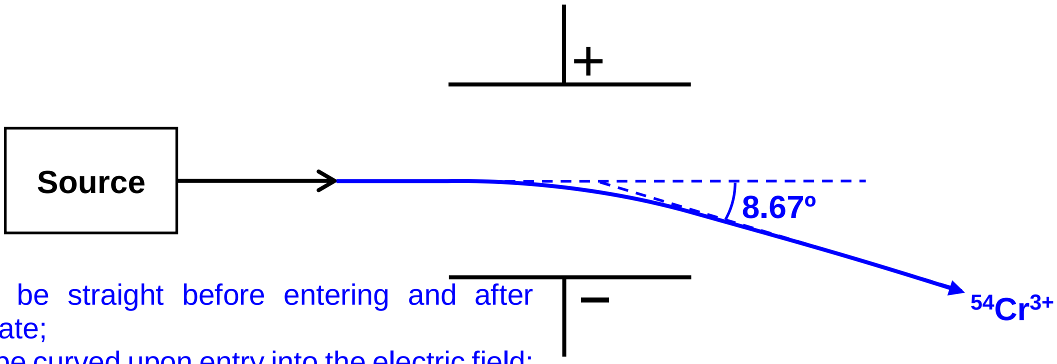
$$\frac{\theta_{^{52}Cr^{2+}}}{\theta_{^{54}Cr^{3+}}} = \frac{q_{^{52}Cr^{2+}}\, m_{^{54}Cr^{3+}}}{m_{^{52}Cr^{2+}}\, q_{^{54}Cr^{3+}}}$$

$$\frac{6}{\theta_{^{54}Cr^{3+}}} = \frac{(+2)(54)}{(52)(+3)}$$

$$\theta_{^{54}Cr^{3+}} = \mathbf{8.67^\circ} \text{ (3 s.f)}$$

[1]

(ii) Sketch clearly how $^{54}Cr^{3+}$ ion will behave when passing through an electric field in the diagram below.



Path should be straight before entering and after exiting the plate;
Path should be curved upon entry into the electric field;
Correct direction of deflection towards the negative plate.

[1]

(d) Chromium can form oxides such as chromium(IV) oxide, CrO_2.

In the presence of dilute sulfuric acid, CrO_2 will disproportionate to Cr^{3+} and $Cr_2O_7^{2-}$.

(i) By constructing the half equations, write the balanced equation for the disproportionation of CrO_2 in acidic solution.

Oxidation: $2CrO_2 + 3H_2O \rightarrow Cr_2O_7^{2-} + 6H^+ + 4e^-$

Reduction: $CrO_2 + 4H^+ + e^- \rightarrow Cr^{3+} + 2H_2O \qquad (\times 4)$

Overall: $\mathbf{6CrO_2 + 10H^+ \rightarrow 4Cr^{3+} + Cr_2O_7^{2-} + 5H_2O}$

[2]

(ii) The resulting solution from **(d)(i)** needed 10.00 cm³ of 0.0200 mol dm⁻³ Fe^{2+} to completely reduce $Cr_2O_7^{2-}$ to Cr^{3+}.

$$Cr_2O_7^{2-} + 6Fe^{2+} + 14H^+ \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$$

Use the data to calculate the mass of CrO_2 that has undergone disproportionation.

$$n(Fe^{2+}) = \frac{10}{1000} \times 0.02 = 0.0002 \text{ mol}$$

$$\frac{n(Cr_2O_7^{2-})}{n(Fe^{2+})} = \frac{1}{6}$$

$$n(Cr_2O_7^{2-}) \text{ present} = \frac{1}{6} \times 0.0002 = 3.333 \times 10^{-5} \text{ mol}$$

$$\frac{n(CrO_2)}{n(Cr_2O_7^{2-})} = \frac{6}{1}$$

$$n(CrO_2) = 6 \times 3.333 \times 10^{-5} = 0.0002 \text{ mol}$$

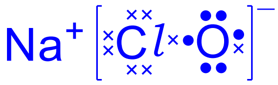
$$\text{Mass of } CrO_2 = 0.0002 \times [52.0 + 2(16.0)] = \underline{0.0168 \text{ g}}$$

[3]

[Total: 10]

2 Hydrazine, N_2H_4, is an inorganic compound largely used as a foaming agent in the preparation of polymer foam. It is manufactured via the Olin Raschig Process where aqueous sodium hypochlorite, $NaC\mathit{l}O$, is mixed with ammonia, NH_3, to produce anhydrous hydrazine.

(a) Draw the dot-and-cross diagram of $NaC\mathit{l}O$.



[2]

A kinetic study has been conducted to determine the order of reaction of the reacting species and the kinetics mechanism.

(b) In one experiment, the following results were obtained.

time / min	$[C\mathit{l}O^-]$ / mol dm^{-3}, where $[NH_3]$ = 0.10 mol dm^{-3}
0	0.0100
30	0.0079
60	0.0062
90	0.0049
120	0.0038
150	0.0030

(i) Using the data above, plot the graph of $[C\mathit{l}O^-]$ against time on the grid in Fig. 2.1. [2]

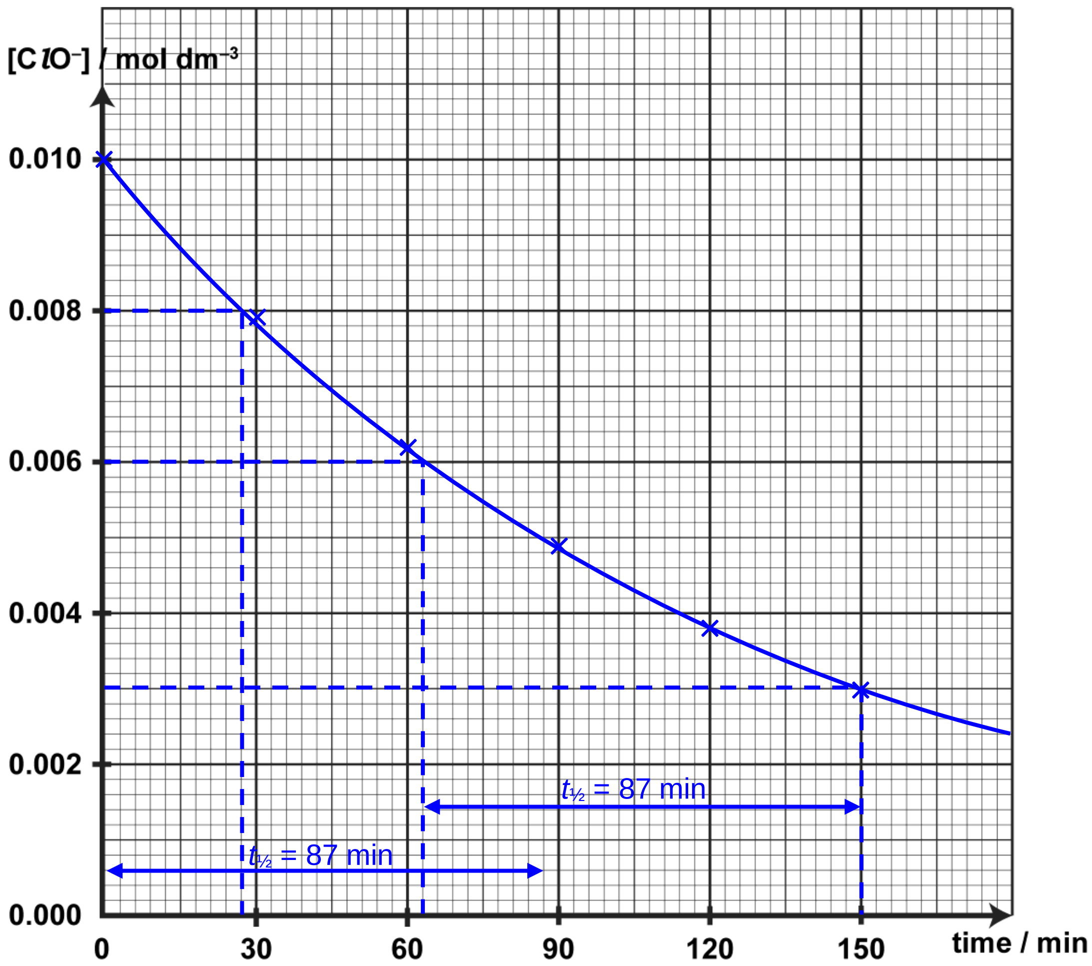


Fig. 2.1

(ii) Use your graph to determine the order of reaction with respect to ClO^-.

From the graph, there is a constant $t_{½}$ = 87 min, thus reaction is first order with respect to ClO^- .

.... [2]

(c) In another series of experiments, the time taken for a fixed volume of hydrazine to be produced was monitored and the following data was obtained.

Table 2.1

experiment	$[ClO^-]$ / mol dm^{-3}	$[NH_3]$ / mol dm^{-3}	time taken / min
1	0.10	0.10	12.5
2	0.10	0.15	8.33
3	0.15	0.05	**16.7**

(i) Deduce the order of reaction with respect to NH_3. Write the rate equation and the units of the rate constant.

By comparing Expt 1 and 2, when **$[ClO^-]$ is kept constant** and **$[NH_3]$ increased by a factor of 1.5** times, the **rate of reaction has increased by 1.5 times**. Thus reaction is **first order with respect to NH_3**.

rate = $k[ClO^-][NH_3]$

unit of k = mol^{-1} dm^3 min^{-1}

or Let rate = $k[ClO^-][NH_3]^x$

$$\frac{\text{rate}_1}{\text{rate}_2} = \frac{\frac{1}{12.5}}{\frac{1}{8.33}} = \frac{k(0.10)(0.10)^x}{k(0.10)(0.15)^x}$$

$$0.6664 = \left(\frac{0.10}{0.15}\right)^x$$

$\Rightarrow x = 1$ (nearest whole number)

Thus order of reaction with respect to NH_3 is 1.

[4]

(ii) Hence, fill in the time taken for experiment 3 to be completed in Table 2.1. [1]

relative rate$_3$ = relative rate$_1$ × 0.5 × 1.5 = 0.0600 mol dm^{-3} min^{-1}

Thus time taken = 1 ÷ 0.0600 = 16.7 min (3.s.f.)

(d) From kinetic studies, the following mechanisms were proposed.

Mechanism A:

$NH_3 + ClO^- \longrightarrow NH_2Cl + OH^-$ (slow)

$NH_3 + NH_2Cl + OH^- \longrightarrow N_2H_4 + H_2O + Cl^-$ (fast)

Mechanism B:

$NH_3 + ClO^- \longrightarrow NH_2Cl + OH^-$ (fast)

$NH_2Cl + NH_3 \longrightarrow N_2H_5^+ + Cl^-$ (slow)

$N_2H_5^+ + OH^- \longrightarrow N_2H_4 + H_2O$ (fast)

(i) Explain which of the two mechanisms is consistent with the kinetic data.

Mechanism A fits the observed kinetic data.

From the rate equation, **1 ClO^- ion and 1 molecule of NH_3 are involved in the slow step**.

[2]

(ii) Hence, write the overall equation for the reaction between sodium hypochlorite and ammonia to produce hydrazine.

$2NH_3 + ClO^- \longrightarrow N_2H_4 + H_2O + Cl^-$ *or* [1]
$2NH_3 + NaClO \longrightarrow N_2H_4 + H_2O + NaCl$

(e) The Olin Raschig Process was modified from the Raschig Process used to produce liquid hydroxylamine, NH_2OH, in the presence of copper metal.

Explain how copper acts as a heterogeneous catalyst in this process.

✓ Copper provides **active sites**
✓ whereby the reactant molecules may be **physically adsorbed**.
✓ This **increases the local concentration of reactants**
✓ And **weakens the covalent bonds** thus lowering the
✓ activation energy for reaction to occur.
✓ The **adsorbed product** molecules **break free** from the catalyst surface and **leave the surface** via desorption.

These processes allow copper to act as a heterogeneous catalyst in this reaction and thus the reaction proceeds at a faster rate via an alternative pathway.

[3]

[Total: 17]

3 Upon strong heating in the absence of air, pyrolysis of chlorodifluoromethane, $CHClF_2$, occurs according to the following equilibrium.

$$2CHClF_2(g) \rightleftharpoons C_2F_4(g) + 2HCl(g) \qquad \Delta H = +128 \text{ kJ mol}^{-1}$$

At 450 °C, a closed vessel with a volume of 18.5 dm^3, contains 1.00 mol of $CHClF_2$. When equilibrium was reached, the amount of $CHClF_2$ in the vessel was found to be 0.200 mol.

(a) Calculate the K_c for the reaction at 450 °C.

	$2CHClF_2(g)$	$\rightleftharpoons$	$C_2F_4(g)$	+	$2HCl(g)$
initial amount/ mol	1.00		0		0
change/ mol	–2(0.400)		+0.400		+2(0.400)
eqm amount/ mol	0.200		0.400		0.800
eqm conc./ mol dm^{-3}	$\frac{0.200}{18.5}$		$\frac{0.400}{18.5}$		$\frac{0.800}{18.5}$

$$K_{c(450^{o}C)} = \frac{[C_2F_4][HCl]^2}{[CHClF_2]^2}$$

$$= \frac{\left(\frac{0.400}{18.5}\right)\left(\frac{0.800}{18.5}\right)^2}{\left(\frac{0.200}{18.5}\right)^2}$$

$$= \underline{\mathbf{0.346\ mol\ dm^{-3}}} \text{ (3 s.f. + units)}$$

[2]

(b) Explain how the equilibrium amount of C_2F_4 would change if the experiment is repeated in a larger vessel at 450 °C.

When a larger vessel is used, the **pressure of the system decreases** and hence, the **equilibrium position shifts right** to **increase the number/amount of gaseous molecules** so as to **increase pressure**.

Therefore, **equilibrium amount of C_2F_4 increases**. [2]

(c) When the reaction was repeated at a different temperature, the K_c of the reaction was found to have a magnitude of 0.780.

(i) Given that the degree of dissociation of $CHClF_2$ is 75%, calculate the equilibrium concentration of HCl at the new temperature.

Let the initial amount of $CHClF_2$ be x.

	$2CHClF_2(g)$	⇌	$C_2F_4(g)$	+	$2HCl(g)$
Initial amount/ mol	x		0		0
Change/ mol	$-0.75x$		$+\frac{0.75x}{2}$		$+0.75x$
Eqm amount/ mol	$0.25x$		$\frac{0.75x}{2}$		$0.75x$
Eqm conc./ mol dm^{-3}	$\frac{0.25x}{18.5}$		$\frac{0.75x}{37.0}$		$\frac{0.75x}{18.5}$

$$K_c = \frac{[C_2F_4][HCl]^2}{[CHClF_2]^2}$$

$$= \frac{\left(\frac{0.75x}{37.0}\right)\left(\frac{0.75x}{18.5}\right)^2}{\left(\frac{0.25x}{18.5}\right)^2} = 0.780$$

x = <u>**4.28 mol**</u> (3 s.f)

equilibrium concentration of $HCl = \frac{0.75x}{18.5}$ = <u>0.173 **mol dm^{-3}**</u> (3 s.f)

[3]

(ii) By considering your answer in **(a)**, state whether the pyrolysis process in **(c)(i)** was repeated at a higher or lower temperature.

<u>**Higher**</u> temperature. .... [1]

[Total: 8]

4 Butane, C_4H_{10}, is a gas that is commonly found in canisters used in portable gas stove.

Some properties of butane are given in Table 4.1 as shown.

Table 4.1

relative molecular mass	58.1
density at 15 ºC	2.48 kg m^{-3}
standard enthalpy change of combustion	–2878 kJ mol^{-1}

(a) Define the term *standard enthalpy change of combustion* of butane.

The **energy released** when **1 mole** of butane is **completely burnt in excess oxygen**

at **1 bar**, at a specified temperature, usually 298 K .... [1]

(b) During winter, a family used a canister containing 225 g of butane to heat up 3 litres of ready-made soup. It took 10 min for the soup to be heated from its initial temperature of 15 ºC to reach its boiling point of 100 ºC.

(i) Calculate the volume, in cm^3, occupied by 225 g of butane at 15 ºC.

$V = 0.225 \div 2.48$
$= 0.0907\ m^3$
$= \mathbf{9.07 \times 10^4}\ cm^3$

[1]

(ii) The volume of the butane canister used is 500 cm^3. Explain the difference between this value and your answer in **(b)(i)**.

Butane in the canister is **pressurised (or compressed)**, and therefore occupies

less volume as pressure is inversely proportional to volume.

[1]

(iii) The combustion process was known to be only 60% efficient. Calculate the mass of butane used to heat the soup from its initial temperature to its boiling point. (Assume specific heat capacity of soup = 4.18 J g^{-1} K^{-1})

Heat absorbed by soup
$= 3000 \times 4.18 \times (100 - 15)$
$= 1.066 \times 10^6$ J

Heat released from combustion
$= 1.066 \times 10^6 \times \frac{100}{60}$
$= 1.777 \times 10^6$ J

Amount of butane
$= 1.777 \times 10^6 \div (2878 \times 10^3)$
$= 0.617$ mol

Mass of butane used to heat the soup
$= 0.617 \times 58.1$
= **35.9 g**

[3]

(iv) Suggest a reason why the combustion of butane is not 100% efficient.

Heat loss to the surroundings/ cooking pot/ canister. *or* ....
Incomplete combustion occurred during the cooking process. *or*
Evaporation of liquid/ soup occurred. *or* Other reasonable answers............... [1]

(c) Under ultraviolet light, butane reacts with bromine to form 2-bromobutane. Describe the mechanism for this reaction.

Name of mechanism: **Free Radical Substitution**

Initiation

$$Br{-}Br \xrightarrow{\text{uv light}} \cdot Br + \cdot Br$$

Propagation

$$Br\cdot + CH_3CH_2CH_2CH_3 \longrightarrow CH_3CH_2\dot{C}HCH_3 + HBr$$

$$Br{-}Br + CH_3CH_2\dot{C}HCH_3 \longrightarrow CH_3CH_2CH(Br)CH_3 + \cdot Br$$

Termination

$$\cdot Br + CH_3CH_2\dot{C}HCH_3 \longrightarrow CH_3CH_2CH(Br)CH_3$$

$$2\ \cdot Br \longrightarrow Br_2$$

$$2\ CH_3CH_2\dot{C}HCH_3 \longrightarrow CH_3CH_2CH(CH_3)CH(CH_3)CH_2CH_3$$

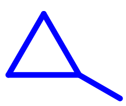
[3]

(d) Butane can be formed by the hydrogenation of but-1-ene, which has a molecular formula of C_4H_8.

(i) State the type of reaction but-1-ene has undergone.

Reduction / addition .... [1]

(ii) Under ultraviolet light, another isomer of C_4H_8, reacts with bromine to form 3 possible mono-brominated products (ignore stereoisomers).
Draw the **skeletal** formula of this isomer.



[1]

[Total: 12]

5 Halogen compounds have a wide range of uses and are commonly used as precursors for conversion into other compounds.

(a) The boiling point of Br_2 is 59 ºC, while the boiling point of I_2 is 184 ºC.

(i) Explain the difference in boiling point between the two halogens.

Iodine has a **larger number of electrons/larger electron cloud** than bromine that leads to a more easily polarised electron cloud, thus forming **stronger instantaneous dipole-induced dipole interactions** (id-id) between molecules. [1]

(ii) Bromine and iodine react to form the interhalogen compound IBr.

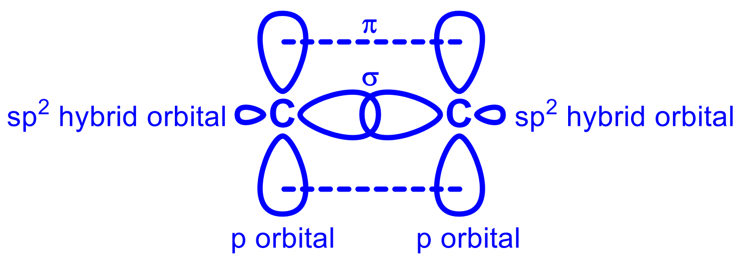
Discuss the relative strength of the different types of intermolecular forces of attraction between IBr molecules, given the boiling point of IBr is 116 ºC.

IBr is a polar molecule that **can form both id-id interactions and permanent dipole-permanent dipole (pd-pd) interactions between molecules**. Since the **boiling point of IBr is in between that of bromine and iodine as the number of electrons in IBr is between that of bromine and iodine**, the **id-id interactions between molecules would be stronger than the pd-pd interactions** between molecules.

.... [2]

(b) When but-1-ene is treated with IBr, the major product, 2-bromo-1-iodobutane, is formed.

(i) Draw a labelled diagram to show how the orbitals overlap to form the C=C bond in an alkene, and state the type of hybridisation involved.



sp² hybridisation

[2]

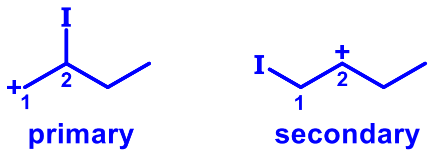
(ii) Name the mechanism of the reaction between IBr and but-1-ene.

Electrophilic addition [1]

(iii) Explain why the major product is 2-bromo-1-iodobutane and not 1-bromo-2-iodobutane, in terms of

- which atom in IBr first reacts with but-1-ene to form the intermediate
- structure of all possible intermediates formed during the reaction.

You may wish to include in your answer the structure of the possible intermediates.

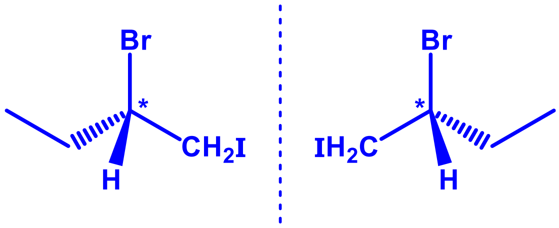


I in IBr has a partial positive charge, such **that $I^{\delta+}$ is the electrophile that first adds across the C=C** to form the intermediate. When the electrophile is **added to the first carbon (C1), a more stable secondary carbocation is formed compared to a primary carbocation if the electrophile was added to C2**. Subsequent addition of Br^- to the carbocation results in formation of 2-bromo-1-butane.

[2]

(iv) State the type of stereoisomerism displayed by 2-bromo-1-iodobutane and draw the structure of the isomers.

Type of stereoisomerism ......Enantiomerism......



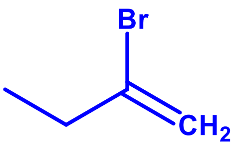
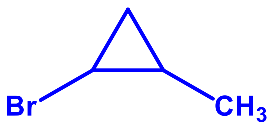
[2]

(c) **A** and **B** are structural isomers with the molecular formula C_4H_7Br.

A is formed from 2-bromo-1-iodobutane and decolourises aqueous bromine.

B does not decolourise aqueous bromine and exhibits stereoisomerism.

Deduce the structures of **A** and **B**. State the reagents and conditions required to convert 2-bromo-1-iodobutane to **A**.

A	**B**
	

Reagents and conditions ......NaOH in ethanol, heat......

[3]

[Total: 13]

Section B

Answer **one** question from this section.

6 (a) State what is meant by the term *dynamic equilibrium*. [2]

It refers to a **reversible process at equilibrium** in which the **rate of the forward reaction equals to the rate of backward reaction** and **rate is non zero**.

(b) Water gas, a mixture of H_2 and CO, can be formed by reacting steam with excess powdered carbon.

$$C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$$

(i) At 1000 K, the equilibrium partial pressure of CO(g) is determined to be 5.40 atm in a sealed vessel of 1 dm^3. Assuming ideal gas behaviour, calculate the minimum mass of carbon required to establish equilibrium under such conditions. [2]

(ii) In another experiment, 1 mol of each gas, $H_2O(g)$, CO(g) and $H_2(g)$ was subjected to high pressure. CO(g) was observed to deviate most from ideal gas behavior. Suggest a reason for the observation. [1]

(i)

$$n_{CO} = \frac{p_{CO}V}{RT}$$

$$= \frac{(5.40 \times 101325)(10^{-3})}{(8.31)(1000)}$$

$$= 0.06584 \text{ mol}$$

Since C ≡ CO, hence minimum mass of C = 0.06584 × 12 = **0.790 g**

(ii) *At high pressure, the volume of the gas particles have more significant effect on the deviation from ideal gas behaviour.*

CO molecule is **larger in size** than H_2O and H_2 molecules. At high pressure and same temperature (300K), the **volume of CO molecules compared to total volume of CO gas/ volume of container is more significant** compared to that of H_2O and H_2 molecules. Hence CO will deviate more from ideality at high pressure.

(c) The CO(g) in water gas can be further reacted in the water gas shift reaction (WGSR).

$$H_2O(g) + CO(g) \rightleftharpoons H_2(g) + CO_2(g) \qquad \Delta H^{\ominus} = -41.1\ \text{kJ mol}^{-1}$$

A sample of water gas containing an equimolar mixture of CO(g) and H_2(g), as well as 0.600 atm of H_2O(g) was introduced to a sealed vessel at 1000 K. At equilibrium, there was 46.2% yield of CO_2 and total pressure was 3.4 atm.

(i) Write an expression for the equilibrium constant, K_p, for this reaction, including the units. [2]

(ii) Assuming that the gases behave ideally, explain why the initial total pressure was 3.4 atm. [1]

(iii) Calculate the equilibrium partial pressures of all species. [3]

(iv) Hence, calculate the value of K_p. [1]

(i) $K_p = \dfrac{(p_{H_2})(p_{CO_2})}{(p_{H_2O})(p_{CO})}$ Unit: no units/ dimensionless

(ii) Since there is **no change in amount of gas** and **amount of gas is directly proportional to pressure of gas**.

Total Initial pressure = Total Equilibrium pressure = 3.4 atm

(iii) Since $p_{total} = p_{H_2O} + p_{CO} + p_{H_2} = 3.4$ atm, the initial partial pressure of CO and H_2 can be calculated.

Given % yield of CO_2 is 46.2%, equilibrium partial pressure of CO_2 can be calculated based on the limiting reagent *i.e.* H_2O.

	H_2O +	CO $\rightleftharpoons$	H_2 +	CO_2
initial partial pressure/atm	0.600	$\frac{1}{2}(3.4-0.600)$ $=1.400$	$\frac{1}{2}(3.4-0.600)$ $=1.400$	0
change in partial	−0.2772	−0.2772	+0.2772	+0.2772
eqm partial pressure/atm	0.3228	1.1228	1.6772	$(\frac{46.2}{100}\times 0.600)$ $=0.2772$

(iv) $K_p = \dfrac{(p_{H_2})(p_{CO_2})}{(p_{H_2O})(p_{CO})} = \dfrac{(1.6772)(0.2772)}{(0.3228)(1.1228)} = 1.283 = 1.28$

(d) Predict, with reasoning, the effect on yield and rate of reaction, if any, for WGSR, when

(i) pressure is increased [2]

(ii) temperature is increased [2]

(i) Since there is **no change in the amount of gaseous molecules**, changes in pressure does not change the position of equilibrium. Hence **no change in yield of CO_2 and H_2.**

At higher pressure, rate of reaction is **faster** as there are **more gas particles per unit volume/ volume decreases causing gas particles are closer together**. Hence frequency of effective collision increases.

(ii) By Le Chatelier's principle, increasing the temperature favours the **endothermic** backward reaction **to absorb some heat**. Hence **yield of CO_2 and H_2 decreases**.

At higher temperature, rate of reaction is **faster** with higher **kinetic** energy. Since there are **more particles with sufficient energy to overcome the activation energy**. Hence frequency of effective collision increases.

(e) Carbon monoxide can also be used to reduce tin(IV) oxide, SnO_2.

$$SnO_2(s) + 2CO(g) \rightarrow Sn(s) + 2CO_2(g) \qquad \Delta H_r^{\ominus}$$

(i) Given:

$$SnO_2(s) + 2H_2(g) \rightarrow Sn(s) + 2H_2O(g) \qquad \Delta H^{\ominus} = +97.2 \text{ kJ mol}^{-1}$$

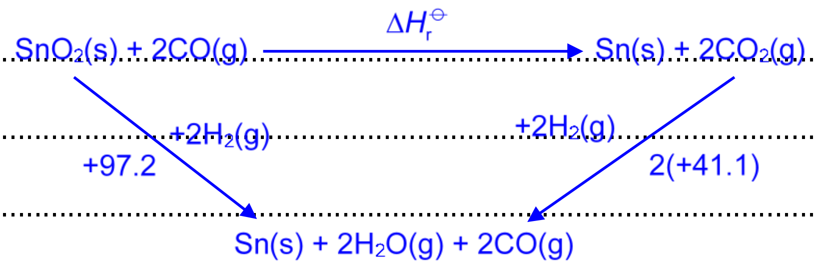
$$H_2(g) + CO_2(g) \rightarrow H_2O(g) + CO(g) \qquad \Delta H^{\ominus} = +41.1 \text{ kJ mol}^{-1}$$

Draw an energy cycle to determine $\Delta H_r^{\ominus}$. [2]

(ii) Calculate Gibbs free energy change for the reduction of SnO_2 by CO at 500 K, given the entropy change is +36.8 J K^{-1} mol^{-1}. [1]

(iii) State how the value of $\Delta G^{\ominus}$ for this reaction changes with increasing temperature. [1]

(i)



$\Delta H_r^{\ominus}$ = 97.2 −2(41.1) = **<u>+15.0 kJ mol⁻¹</u>**

(ii) $\Delta G^{\ominus} = \Delta H^{\ominus} - T\Delta S^{\ominus}$ = +15000 – 500(36.8) = **<u>–3400 J mol^{-1}</u>**

(iii) With increasing temperature, **<u>$-T\Delta S$</u>** term becomes **<u>more negative</u>** or **<u>$T\Delta S$</u>** term becomes **<u>more positive.</u>** This makes **<u>ΔG more negative since ΔH is positive</u>**. Hence, the reaction is **<u>spontaneous at high temperatures</u>**.

[Total: 20]

7 Alkanes can be synthesised by passing an electric current through an aqueous solution of the sodium salts of two different mono-carboxylic acids, RCO_2H and $R'CO_2H$.

An example of such a synthesis is shown.

$$\underset{\text{salt of carboxylic acid } \mathbf{A}}{RCO_2^-(aq)} + \underset{\text{salt of carboxylic acid } \mathbf{B}}{R'CO_2^-(aq)} \longrightarrow \underset{\mathbf{C}}{R'{-}R'} + \underset{\mathbf{D}}{R{-}R'} + \underset{\mathbf{E}}{R{-}R}$$

mixture of alkanes **C**, **D** and **E**

(a) Explain, in terms of structure and bonding, the following observations.

(i) The acid RCO_2H is a liquid at room temperature with a melting point of –47 °C, while its sodium salt, $RCO_2^-Na^+$, melts at 320 °C. [2]

(ii) The acid RCO_2H is less soluble than its sodium salt, $RCO_2^-Na^+$, in water. [2]

(i) **$RCO_2^-Na^+$ consists of giant ionic structure while RCO_2H consists of simple molecular structure. More energy is required to break the stronger electrostatic forces of attraction between the oppositely charged ions in $RCO_2^-Na^+$ as compared to the intermolecular hydrogen bonding in RCO_2H.** Hence $RCO_2^-Na^+$ melts at a much higher temperature than RCO_2H.

(ii) **RCO_2H is able to form intermolecular hydrogen bonds with water molecules** while **$RCO_2^-Na^+$ is able to form stronger/ more favourable ion-dipole interactions with water molecules.**

(b) On titration, a solution containing 0.100 g of the acid **A**, RCO_2H, required a volume of 11.4 cm^3 of 0.100 mol dm^{-3} NaOH to reach end point. Show that the molar mass of acid **A** is 87.7 g mol^{-1} and hence suggest the two possible structures of acid **A**. [3]

$RCO_2H + NaOH \rightarrow RCO_2Na + H_2O$

Amount of NaOH = (11.4/1000)(0.100) = 1.14×10^{-3} mol

Amount of RCO_2H = 1.14×10^{-3} mol

Molar mass of **A** = $0.100/(1.14 \times 10^{-3})$ = 87.7 g mol^{-1}

Hence **A** is either **$CH_3CH_2CH_2CO_2H$** or **$(CH_3)_2CHCO_2H$**

(c) A gaseous sample of 0.20 g of alkane **D** occupies a volume of 8.70×10^{-5} m^3 at a temperature of 380 K and a pressure of 1.01×10^5 Pa. Use these data to calculate the molar mass of alkane **D** and hence suggest its molecular formula. [2]

Assume **D** is an ideal gas and apply the ideal gas equation.

$$pV = \frac{m}{M}RT$$

$$M = \frac{mRT}{pV} = \frac{0.20 \times 8.31 \times 380}{1.01 \times 10^5 \times 87 \times 10^{-6}} = 71.9 \text{ g mol}^{-1}$$

Since **D** is an alkane, **D** has molecular formula, C_5H_{12}.

(d) Use your answer in **(b)** and **(c)** to give

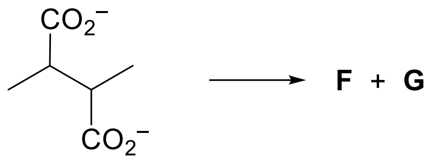
(i) the structure of acid **B**, and [1]

(ii) one possible structure of alkane **E**. [1]

(i) Since **A** is $CH_3CH_2CH_2CO_2H$ or $(CH_3)_2CHCO_2H$,
R = $CH_3CH_2CH_2$ or $(CH_3)_2CH$
Since **D** is C_5H_{12}, R′ = CH_3CH_2,
So **B** is **$CH_3CH_2CO_2H$**.

(ii) Alkane **E** is R–R and can be **$CH_3(CH_2)_4CH_3$** or **$(CH_3)_2CHCH(CH_3)_2$**

(e) When the sodium salt of butane-2,3-dioic acid is treated in a similar way by passing through an electric current, two gases, **F** and **G** are obtained.



Both **F** and **G** are isomers and they form butane upon reaction with hydrogen gas over a palladium catalyst.

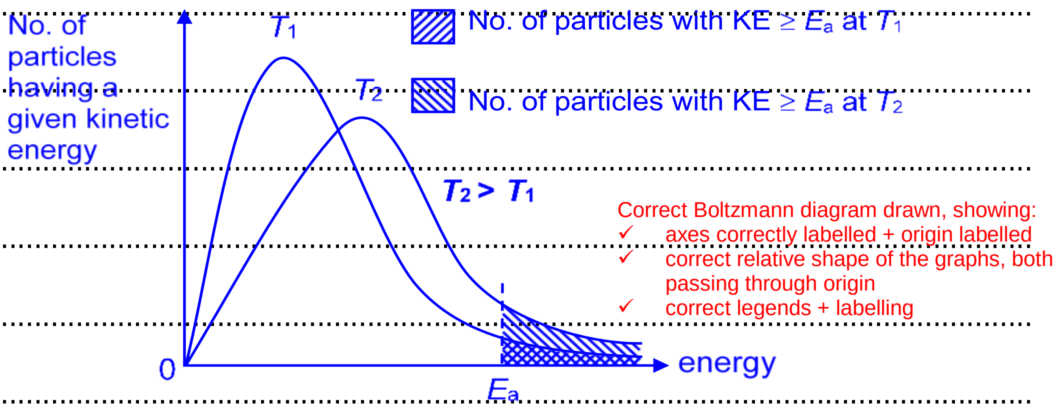
(i) Draw the **skeletal** formulae of **F** and **G** and name them. [3]

(ii) The reaction with hydrogen gas takes place faster at elevated temperatures. With an appropriate sketch of the Boltzmann distribution, explain why this is so. [3]

(i)

***trans*-but-2-ene** ***cis*-but-2-ene**

(ii)



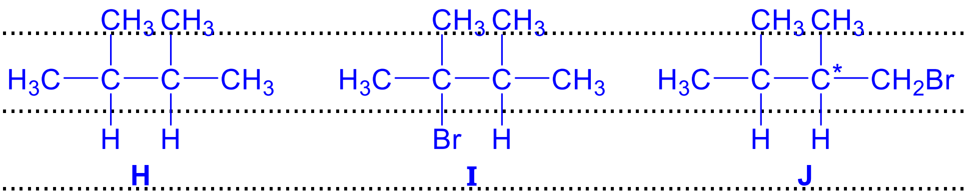
As temperature increases, **average kinetic energy** of the particles increases and **number of particles with energy equal to or greater than E_a increases**. Thus **frequency of effective collision increases**, and the **rate constant** and **rate of reaction increases** when the temperature is increased.

(f) In another separate experiment, a new alkane **H**, C_6H_{14}, was produced. When reacted with bromine under ultraviolet light, **H** produced only two isomeric monobromo compounds **I** and **J**, with the formula $C_6H_{13}Br$. Compound **J** was chiral. Draw the structures of **H**, **I** and **J**. [3]

H, C_6H_{14}, underwent free radical substitution with Br_2 under ultraviolet light, to produce only two isomeric monobromo compounds **I** & **J**.

⇒ **F** contains only two types of H atoms.

Compound **J** was chiral.⇒ **J** has a chiral carbon atom.



[Total: 20]